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Treatment of Oil-Based Produced Water and Spent Caustic Wastewater Using Advanced Electrochemical Technology

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Abstract

Oil-based produced water (PW) and spent caustic (SC) are challenging industrial waste streams generated from petroleum and petrochemical industries. Their complex physiochemical nature, and the recent strict Saudi discharge regulations necessitate advanced treatment methods. This study aims to evaluate the treatment of both waste streams using a batch-mode pilot-scale advanced electrochemical system (AOT). The objective includes optimizing COD removal by investigating the effects of key operating parameters including current density, pH, and air flow rate. In this study, synthetic PW and pretreated authentic ethylene plant SC were treated in a two-separate set of experiments. The PW used in this study had a total dissolved solids (TDS) concentrations of 16,000 mg/L and organic carbon (COD) concentration of 250 mg/L, while SC has TDS of 23000 mg/L and COD of 670 mg/L. The AOT unit consists of anodic oxidation (AO) and in-situ generation of strong oxidants (i.e. hydrogen peroxide) in the electrolytic cell. The electro-cell is outfitted with a boron-doped diamond (BDD) anode and gas diffusion (GDE) cathode. The investigation verified the influential role of diverse operational factors in the treatment process. The results showed that for the PW, a TOC removal of 84% was achieved at pH 12, 200 mA/cm² electric current density and a circulation rate 0.2 L/min within 4 hours. The results of the SC treatment part showed a maximum COD removal of 90% at pH 4, 150 mA/cm² electric current density and circulation rate 0.2 L/min within 3 hours. It was found that adding iron and air did not lead to any considerable improvement of the COD reduction of both PW and SC. This work clearly demonstrated that electrochemical post treatment is an efficient, cost-effective and environmentally friendly technique for the treatment of both pretreated spent caustic and oil-based produced water.

Keywords: Advanced electrochemical treatment, Produced water, Spent caustic, BDD

Introduction

Industrial wastewater management in the petroleum and petrochemical industries remains a persistent environmental challenge (Singh et al., 2023; Villarín & Merel, 2020). The release of huge and inadequately treated wastewater into the environment can significantly contaminate ecosystems and pose risks to human

health. Strict discharge standards have driven industries to seek efficient, cost-effective technologies for wastewater treatment, ensuring permissible and safe environmental disposal levels ([Abdelfattah & El-Shamy, 2024](#); [Dutta et al., 2021](#)). Oil-based produce water (PW) and spent caustic (SC) are examples of such wastes.

Produced water (PW) represents the primary waste stream generated by the oil and gas industry, resulting from both onshore and offshore crude oil and natural gas production ([Zhang et al., 2017](#)). PW constitutes the largest volume of wastewater, amounting globally to approximately 250 million barrels daily, compared to around 80 million barrels of oil produced, indicating a water-to-oil ratio of approximately 3:1 ([Dawoud et al., 2021](#); [Shahbaz et al., 2023](#)). By 2020, global PW production reached roughly 340 billion barrels ([Sobolciak et al., 2020](#)). Due to its large volume and hazardous content, including toxic organic and inorganic contaminants, PW poses significant environmental threats, potentially affecting aquatic ecosystems, groundwater, and human health ([Hedar & Budiyo, 2018](#)). Similarly, spent caustic (SC) is another significant industrial wastewater, presenting substantial treatment and disposal challenges for petrochemical and refining industry ([Alipour & Azari, 2020](#); [Rita et al., 2020](#); [Shokri & Nasernejad, 2023](#)). SC originates predominantly from oil refineries and olefin production plants ([Davarnejad & Bakhshandeh, 2018](#); [Rita et al., 2020](#)), used primarily to remove contaminants such as sulfides, carbonates, and organic compounds from process streams ([Ben Hariz et al., 2013](#); [Delnavaz et al., 2021](#)). Typically, SC is hazardous due to its corrosive nature (high pH), toxicity (presence of H_2S or phenols), high chemical oxygen demand (COD), and strong odors (high sulfide content). Typical spent caustic from ethylene plants may contain up to 6% NaHS, 5% $NaHCO_3$, and exhibit COD values often in tens of thousands of $mg\ O_2/L$ ([Alipour & Azari, 2020](#); [Maugans et al., 2010](#)).

Conventional chemical, and biological treatment technologies for PW and SC often face significant limitations. These limitations include pollutant phase transfer, high operational costs, generation of toxic by-products and sludge, limited pollutant removal efficiency, and susceptibility of biological treatments to shocks due to high salinity and toxicity ([Gul Zaman et al., 2021](#); [Mahmodi et al., 2022](#)). Therefore, searching for a most efficient and cost-effective technologies is the focus of many researchers and engineers. Advanced oxidation process (AOP) process which relies on the generation of a very strong oxidant such as hydroxyl radicals (OH^*).

Numerous studies have demonstrated the efficiency of AOP methods using BDD and other electrodes for treating PW and petrochemical wastewater. For example, electrooxidation processes using BDD electrodes have achieved substantial COD and organic pollutant removal efficiencies, outperforming other electrode materials such as PbO_2 and Pt ([Gargouri et al., 2014](#); [Rocha et al., 2012](#)). Additionally, pilot-scale studies employing BDD electrodes have confirmed their effectiveness in degrading difficult-to-treat organic compounds, including phenolics, hydrocarbons, and BTEX compounds, under optimized conditions ([Abou-Taleb et al., 2021](#); [Tawabini et al., 2020](#); [Yaqub et al., 2017](#)).

On the other hand, SC is frequently treated by dilution combined with biological methods, deep-well injection, incineration, humid peroxide oxidation (HPO), and wet air oxidation (WAO). However, these methods have drawbacks such as high costs, partial effectiveness, and limited capacity for COD reduction ([Alnaizy, 2008](#); [Suarez, 1996](#)). To overcome these limitations, advanced electrochemical oxidation methods have emerged as promising technologies for wastewater treatment, capable of effectively treating persistent organic and inorganic pollutants ([Chaplin, 2019](#); [Hassani et al., 2022](#)). Electrochemical oxidation (EO), particularly anodic oxidation, is widely acknowledged as an efficient advanced oxidation process (AOP) due to its simple installation, minimal chemical usage, and potential for complete mineralization of contaminants into harmless products like CO_2 and water ([AlJaberi et al., 2023](#)). Among available electrodes, boron-doped diamond (BDD) electrodes stand out for their excellent corrosion resistance, broad potential window, high conductivity, and chemical stability, making them ideal for the mineralization of persistent pollutants ([Einaga, 2022](#)).

Despite the above studies, there is still a research gap presented by the lack of identifying the optimum operation of AOP using a pilot-scale unit for the specific source of waste streams investigated in this work. Therefore, this study aims to evaluate and optimize the electrochemical treatment of two waste streams synthetic produced water and authentic spent caustic using boron-doped diamond (BDD) anodes at the pilot scale. The effects of current density, pH, and airflow on the removal efficiencies of total organic carbon TOC and chemical oxygen demand COD removal were studied. In addition, reaction kinetics were analyzed and fitted to the pseud-first and pseud-second order models.

Materials and Methods

Synthetic Produced Water (PW)

The synthetic PW was prepared following the procedure suggested by Dardor et al. (Feng et al., 2022) with slight adjustments. First, brine was prepared by dissolving different salts listed in Table 1 into one liter of deionized water. Then, 1000 mL of this prepared brine was transferred into a 2 L beaker. Subsequently, 150 mg of sodium dodecyl sulfate (SDS) and 730 mg of a mixture containing three crude oil types, as detailed in Table 2, were added to the brine. This provided a 5:1 oil-to-surfactant weight ratio. The resulting solution was stirred at 1000 rpm for 30 minutes, followed by sonication for an additional 30 minutes to ensure emulsion stability. Afterward, the mixture was transferred into a separatory funnel and left undisturbed for 4 hours to allow any free oil layer to separate. Finally, the aqueous phase was transferred into a glass bottle, and the volume was adjusted to 2.5 L, preparing it for treatment in the electrochemical pilot system. The characteristics of the prepared synthetic produced water (PW) are shown in Table 3 below.

Table 1—Chemical compounds added to prepare the brine in 1.0 L volume

Compound	Weight (g)	Compound	Weight (g)
Sodium carbonate	0.05	Magnesium Chloride	5.27
Sodium Chloride	141.24	Potassium Chloride	3.22
Sodium Sulfate	0.24	Barium Chloride	0.01
Calcium Chloride	26.81	Strontium Nitrate	1.09
Phenol	0.8		

Table 2—Properties and quantities of crude oil samples and surfactant used for PW synthesis

Component	Description	Volume (mL)	density (g/mL) @ 25 Co	Weight (mg)
Crude 1	Medium	0.3	0.84	252
Crude 2	Arabian Light	0.3	0.814	244
Crude 3	Condensate	0.3	0.77	231
Surfactant	SDS	-	-	150

Table 3—Characteristics of the synthetic PW

Parameter	Value	Parameter	Value
Chloride (Cl ⁻) mg/L	6301.06	Strontium (Sr ²⁺) mg/L	0.55
Sodium (Na ⁺), mg/L	3776.44	TDS mg/L	16,200
Calcium (Ca ²⁺), mg/L	277.05	pH	8.3
Magnesium (Mg ²⁺), mg/L	73.10	Conductivity (μS/cm)	24,320
Sulfate (SO ₄ ²⁻), mg/L	160.94	TOC (mg/L)	250
Potassium (K ⁺), mg/L	61.02	Turbidity (NTU)	2.49
Bicarbonate (HCO ₃ ⁻) mg/L	41.40	IC (mg/L)	6.25
Bromide (Br ⁻) mg/L	3.90	UV@254 nm	0.33

Authentic Spent Caustic (SC)

The Spent Caustic used was obtained from an ethylene plant located in the Eastern Province of Saudi Arabia near Dammam City. Spent caustic samples were obtained after being treated by Wet Air Oxidization (WAO) in order for further treatment by electrochemical pilot scale plant. The characteristics of the authentic ethylene-based SC used in this study are shown in Table 4. The initial COD was 670 mg/L and the target of this study is to reduce the COD to a value close to 50 ppm to meet the allowable discharge level.

Table 4—Spent Caustic Characteristics Used in the Study

Parameters	Level	Unit	Parameters	Level	Unit
COD	670	(mg/l)	Ammonia	1.7	(mg/l)
TOC	251	(mg/l)	Sulfates (SO ₄)	28191	mg/L
pH	10	-	Turbidity	9.66	NTU
E. Conductivity	36,000	uS/cm	Bicarbonate alkalinity	527.36	mg of CaCO ₃ /L

Advanced Electrochemical Treatment (AOT) Pilot System

In this study, a laboratory-scale Electrochemical Treatment System (EST) was employed to treat both PW and SC. The ETS (Fig. 1) uses an Electro MP Cell from ElectroCell Co., Denmark, a modular plate-and-frame electrochemical cell that can operate in either divided or undivided mode. The cathode used is a carbon-PTFE GDE, and the anode is a BDD electrode; each electrode has an effective area of 0.01 m². The tests were conducted using a constant-current power supply from Delta Electronika (SM 3300-series, Netherlands), capable of providing a maximum current of 22 A and voltage of 70 V. Operational parameters such as conductivity, pH, temperature, pressure, and oxidation-reduction potential (ORP) were continuously monitored by sensors at both inlet and outlet points of the ETS unit. In the treatment run, a 2.5 L of the wastewater was circulated through the cell using a horizontal multistage pump, controlled by an inverter, which has a maximum flow capacity of 2 m³/h at 50 m head. The liquid flow rate was monitored using a float-type flow meter.

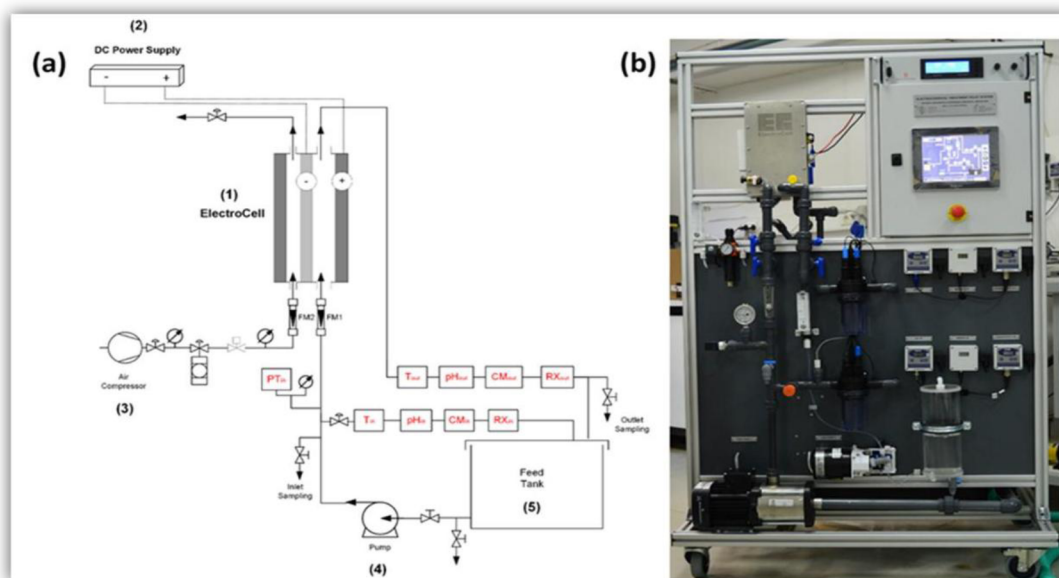


Figure 1—Electrochemical treatment pilot system (a) schematic diagram and (b) photograph of the actual system

Analytical methods

Several analytical methods were utilized to measure parameters during the study. The effectiveness of the PW treatment process was primarily indicated by the TOC removal rate. TOC measurements were conducted using an Analytica Jena multi N/C 3100 TOC analyzer fitted with an auto-sampler. Conductivity and pH were assessed using a PCD 650 conductivity meter. Other parameters, including alkalinity, turbidity, and conductivity, were analyzed comprehensively using a MANTECH PC-BOD multi-parameter analyzer. Concentrations of cations and anions were quantified using a dual-column Metrohm 850 Professional IC. For SC, the primary indicator was COD, measured using a HANNA HI 83224 wastewater treatment photometer coupled with a HANNA 839800 COD reactor. To calibrate the instrument, a stock COD solution (850 mg potassium hydrogen phthalate diluted to 1000 mL with deionized water) was prepared. COD calibration solutions (0 to 1000 ppm) were measured, and readings were recorded.

Experimental procedure

The pilot ETS was flushed thoroughly with water before initiating the PW treatment experiments. After flushing, the system was completely drained and refilled with either PW or SC samples. This step was consistently repeated after every experiment to maintain result accuracy. To ensure the solution was homogeneous, the feed was recirculated inside the electrochemical cell at a flow rate of 0.2 m³/h. Once pH and conductivity readings stabilized at both the inlet and outlet (marking time zero), the power supply was activated at the preset current, and the treatment began. After each experiment, physicochemical properties (pH, e. conductivity, and TOC) of samples collected at specific intervals (0, 30, 60, 120, 180, 210, and 240 minutes) were recorded. For TOC measurement in PW samples, 5 mL of each sample was placed into a 40 mL vial and diluted to a total volume of 40 mL, achieving an 8-fold dilution. After thoroughly mixing to ensure uniformity, the sample was analyzed using the TOC analyzer.

For SC samples, approximately 50 mL of solution was collected to measure the initial COD concentration. Throughout the batch experiments, various operational parameters including current density (A/cm²), pH, wastewater circulation rate (m³/h), and airflow rate (NL/min) were adjusted to determine optimal conditions.

The removal efficiency of TOC or COD was calculated using Eq. 1.

$$TOC \text{ or COD removal } (\%) = \frac{C_0 - C_t}{C_t} * 100 \quad (1)$$

Where C_o and C_t are initial and concentration at time t of TOC or COD.

The average current efficiency (% CE) for each treated solution at a particular time (t) was computed using Eq. 2 (Feng et al., 2022), where C_o and C_t are the initial and concentration at time t of TOC or COD (g/L^{-1}), F is the Faraday constant 96487 mol^{-1} , I is the current in Amps, V is the volume of solution (L), t is the reaction time.

$$\%CE = \frac{C_o - C_t}{8 * I * t} F * V * 100 \quad (2)$$

The energy consumption per unit of TOC or COD mass at a particular time was calculated using Eq. 3 (He et al., 2022), where E_{cell} is the supply voltage (V), t is electrolysis time (h), ΔC is the change of TOC or COD concentration (mg/L^{-1}), V is the volume (L) and I is the current (A).

$$EC = \frac{E_{\text{cell}} * I * t}{\Delta C * t * V} \quad (3)$$

Results And Discussion

Treatment of Produced Water (PW)

A preliminary control experiment was conducted without applying any electric current, to evaluate if there were TOC losses due to adsorption or volatilization while the wastewater was being recirculated in the pilot setup. As shown in Figure 2, after 4 hours of continuous recirculation, the measured losses were minimal. The TOC concentration remained almost unchanged, with just around a 13% decrease, demonstrating that the system performed with good stability.

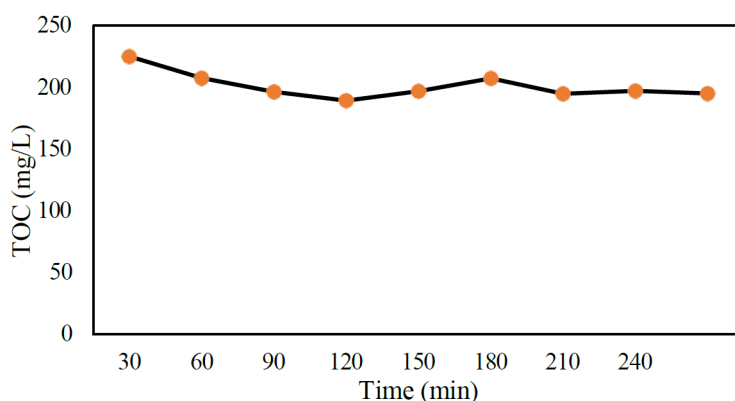


Figure 2—TOC levels during circulation in the pilot system with no airflow and no current.

Effect of Operational Parameters on TOC removal. The synthetic PW was treated under moderate operational conditions (airflow rate: 2 NL/min, pH: 7, current density: 125 mA/cm²) for a total duration of 500 minutes, to study the effect of electrooxidation time on TOC removal, as presented in Figure 3 (a). The results showed a gradual increase in TOC removal, reaching approximately 75% in 240 minutes. Beyond this duration, extending the treatment time had minimal influence on further TOC removal. This limited improvement could be due to the applied current exceeding the limiting current, suggesting that the process was under mass transport control. As a result, current efficiency decreases, possibly because of electrode polarization or the generation of oxygen. Under these conditions, the TOC removal follows an exponential pattern (Panizza et al., 2008). Based on this observation, all further experiments in this study were carried out for 240 minutes, which was found to be sufficient to achieve significant TOC removal.

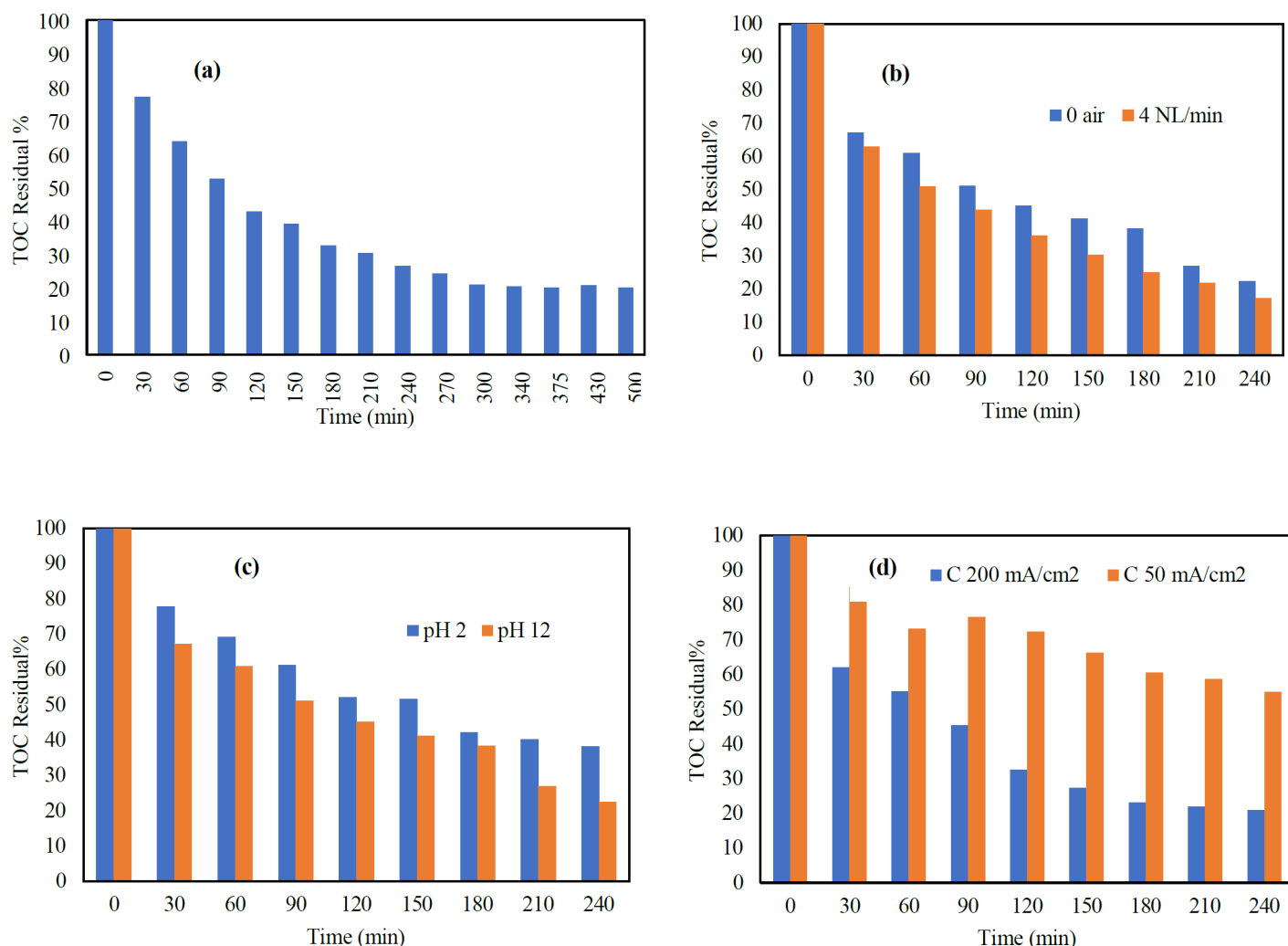


Figure 3—Effects of various operational parameters on TOC removal efficiency: (a) Effect of contact time, (b) Effect of airflow rate, (c) Effect of pH, and (d) Effect of current density

The effect of air flow rate on TOC removal is shown in Fig. 3 (b). Supplying air at the GDE cathode can improve the electrooxidation efficiency in undivided cells due to hydrogen peroxide (H_2O_2) production at the electrode surface (Salmerón et al., 2019). To investigate the effect of airflow on TOC removal from PW, air was introduced to the cathode's air chamber at three different flow rates: 0, 2, and 4 NL/min. Figure 3 (b) shows how TOC removal changed over time under these airflow rates. It was clearly observed that increasing the airflow from 0 to 4 NL/min slightly enhanced the TOC removal efficiency. This experiment was carried out at pH 12 and a current density of 125 mA/cm². The slight improvement in TOC removal with increased airflow can be linked to the additional oxygen delivered into the system. This oxygen enhances oxidation reactions necessary for TOC removal by generating H_2O_2 at the cathode, which subsequently leads to the formation of hydroxyl radicals ($\bullet OH$) at the anode. Thus, organic compounds are more effectively degraded via a combined advanced oxidation/ H_2O_2 mechanism as illustrated by Eqs. 4 and 5.

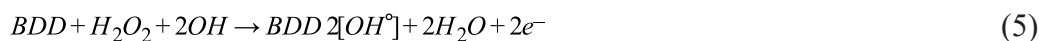


Figure 3 (c) illustrates how changing pH affects TOC removal under different experimental conditions. The results clearly show that increasing the pH from 2 to 12 consistently enhances TOC removal. Over the 240-minute treatment period, TOC removal improved from approximately 60% at pH 2 to about 75% at pH

12. This increase can be explained by several factors. During electrooxidation at the BDD anode, highly reactive hydroxyl radicals ($\bullet\text{OH}$) are formed. At higher pH levels, the higher concentration of hydroxide ions promotes the formation of additional hydroxyl radicals, which strongly oxidize organic pollutants in PW. Additionally, at elevated pH, the surface of the electrode becomes more negatively charged due to the abundance of hydroxide ions (OH^-). This negative charge helps attract organic molecules toward the electrode surface, thus enhancing their adsorption onto the BDD anode and further improving TOC removal efficiency.

Figure 3 (d) shows how electric current influences TOC removal. The results clearly demonstrate a consistent improvement in TOC removal as the electric current increases. Specifically, at pH 7, The results indicate a significant enhancement in TOC removal from around 45% to about 80% as current density rises from 50 to 200 mA/cm^2 . This improvement can be explained by the increased generation of reactive oxygen species (ROS), especially hydroxyl radicals ($\bullet\text{OH}$), known for their strong oxidative power. The enhanced oxidative conditions facilitate the decomposition of complex organic contaminants in PW, resulting in greater TOC removal, as indicated by Eq. 6. Additionally, higher current accelerates the oxidation process, leading to more effective decomposition of organic substances. Moreover, the chemical stability and superior conductivity of the BDD anode ensure efficient electrode surface activation, further improving the electro-oxidation efficiency.



Kinetics and Current Efficiency (CE) for PW Treatment. The data were fitted to different kinetic models, including pseudo-first and pseudo-second order models using Eqs. 7 and 8, as shown in Figure 4 (a) and (b), which shows that at a lower current density of 50 mA/cm^2 , the TOC removal followed a pseudo-second-order model. In contrast, when the current was increased to 200 mA/cm^2 , the behavior shifted to follow a pseudo-first-order model. This difference suggests that at lower currents, the generation of hydroxyl radicals is limited, which may reduce the efficiency of TOC degradation.

$$\ln \frac{C_0}{C_t} = k_1 t \quad (7)$$

$$\frac{1}{C_t} - \frac{1}{C_0} = k_2 t \quad (8)$$

On the other hand, Figure 4 (c) illustrates the current efficiency calculated at various times and current densities using Eq. 2. Initially, a rapid reduction in TOC was observed at the beginning of the treatment. However, achieving the maximum TOC removal of around 84% required approximately 4 hours. This trend could be explained by mass transfer phenomena. At the start of the treatment, the TOC concentration near the electrode is high, leading to faster removal rates and consequently higher current efficiency. As treatment proceeds, the TOC concentration decreases, slowing down the mass transfer and reducing the current efficiency. Additionally, it was observed that increasing current density significantly lowered the current efficiency. Similar trends were reported previously in literature, suggesting that secondary reactions such as polarization effects and oxygen evolution are responsible for this decline (Brillas et al., 2009; Xing et al., 2018).

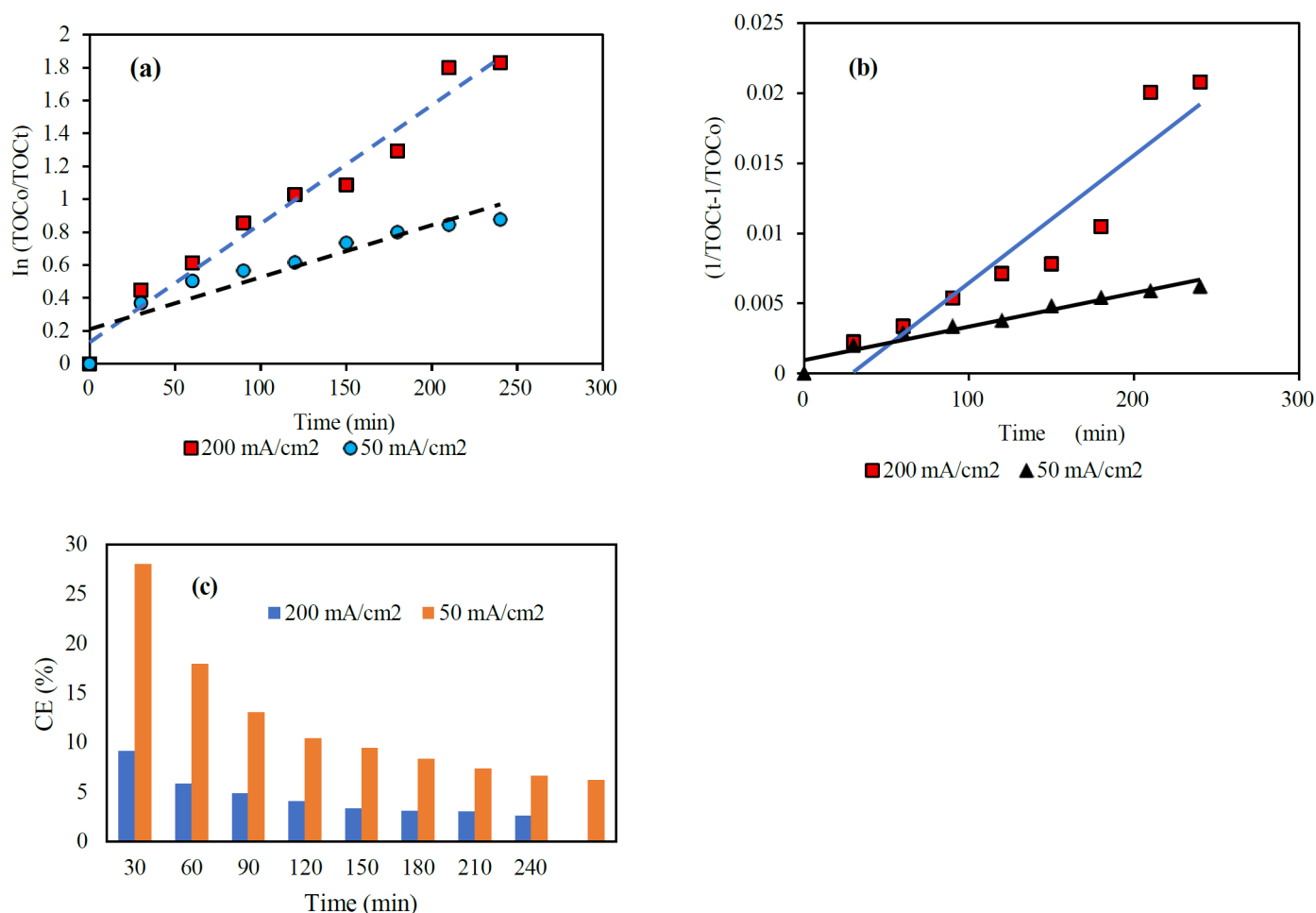


Figure 4—Kinetic models under varying conditions;(a & b) Pseudo-first order and pseudo-second-order kinetic models; (c) current efficiency for TOC removal

Treatment of the Authentic Spent Caustic (SC)

Effect of Operational Parameters on COD removal. A preliminary control experiment was performed without applying an electric current to determine whether COD changes could occur due to adsorption or volatilization during wastewater recirculation within the pilot system. Figure 5 (a) shows that after 3 hours of continuous recirculation, the measured COD losses were minimal.

The effect of pH on the removal of COD was investigated by carrying out three experiments under acidic (pH 4), neutral (pH 7) and alkaline (pH 10) conditions of the spent caustic. The pH was adjusted by using 98% sulfuric acid. As shown in Figure 5 (b), the best % COD removal after 3 hours was around 91% at pH 4 compared to the pH at 7 and 10. The results show that the BDD oxidation of the spent caustic solution can take place in a wide pH range, since the efficiency of the process is not dramatically affected by the pH. The better efficiency at acidic pH is probably attributed to the speciation of the dissolved organics (which may differ with pH) and the formation of sulfate radicals (due to the addition of the sulfuric acid) which can synergistically oxidize the organics. Considering the larger COD removal and the higher current efficiency, pH 4 was selected as the optimum for the conduction of the next series of experiments.

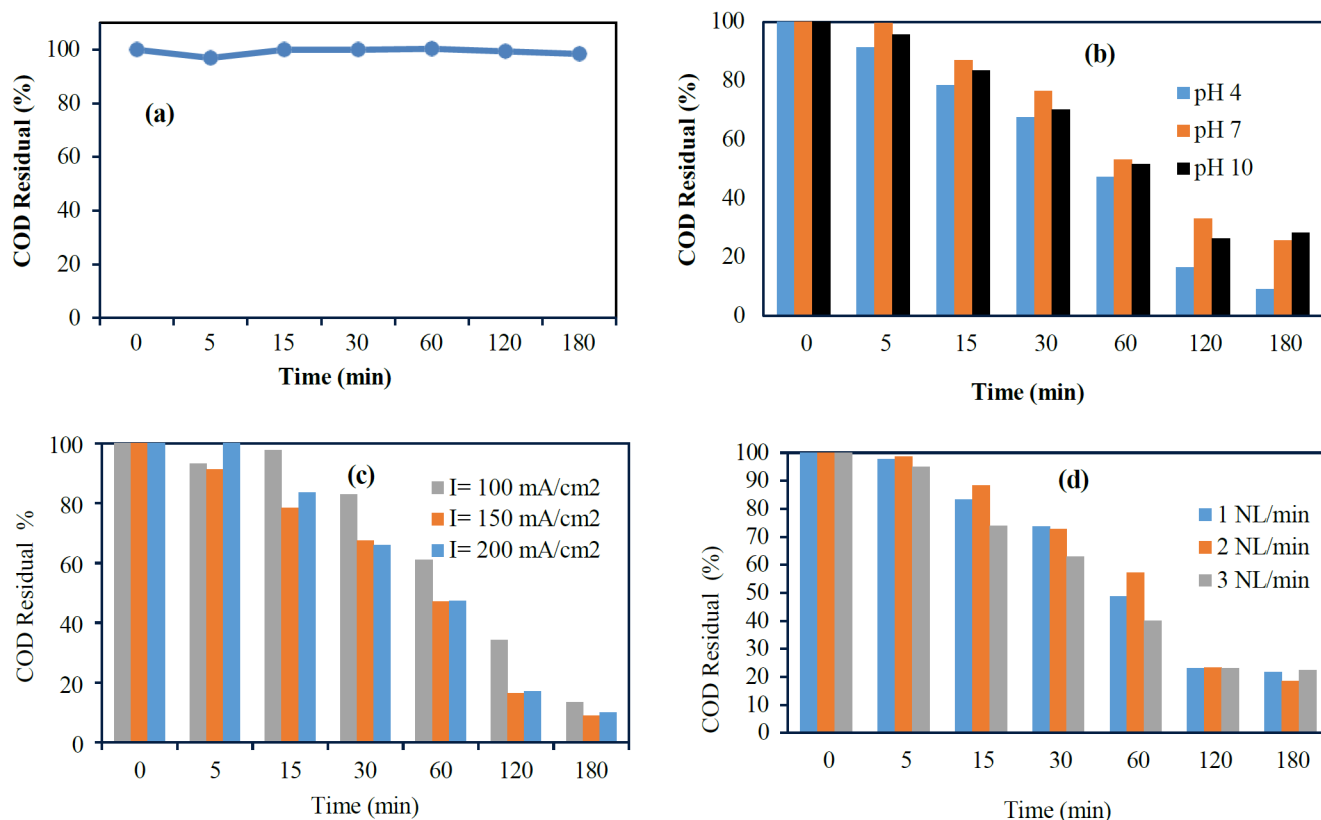


Figure 5—Effects of various operational parameters on COD removal efficiency: (a) Effect of contact time, (b) Effect of pH, (c) Effect of current density, (d) Effect of airflow rate

The effect of applied electric current on COD removal studied at pH 4 is depicted in Figure 5 (c). The results that have been obtained at 150 and 200 mA/cm² were very similar, showing that the system has already reached its maximal oxidation power at around 150 mA/cm². This pattern is consistent with literature since parasitic reactions like oxygen evolution can cause current efficiency to decrease at high applied currents. Figure 5 (c) shows that around 91% of organic compounds been oxidized in the cell after 3 hours due to the high oxidation power of the heterogeneously generated hydroxyl radicals (BDD (·OH)), as shown in Eq. 9. (Brillas et al., 2009)



It was noted that COD removal reached close to 90% at the 3 currents dosages used after contact time of 3 hours. However, the removal of COD using 100 mA/cm² was lower than the other 2 dosages. Therefore, the 150 mA/cm² current value was selected as the optimum current dosage to lower the cost of the energy needed to run the unit.

The effect of air flow on COD removal is shown in Figure 5 (d). In this experiment, pH, applied electric current, and circulation rate were fixed at 4, 15 mA/cm² and 0.2 m³/hr, respectively. Airflow rate varied at 1, 2 and 3 (NL/min) respectively, by blowing compressed air into the GDE electrode's air chamber. The system's oxidation power can be significantly improved by an excessive air supply due to the increased electro-generation of hydrogen peroxide (H₂O₂) (Salmerón et al., 2019). As shown in Figure 5 (d), the COD decay was similar without and with air feeding. This result demonstrates that excess air may not always enhance the combined electrochemical treatment's ability to treat spent caustic. This may be due to the excess oxygen in the bulk prevents the direct anodic oxidation of the intermediate organic compounds.

Kinetics and Current Efficiency (%) for SC Treatment. The reaction kinetics was studied under the best conditions (i.e., pH = 4, I = 150 mA/cm², circulation flow rate = 0.2 m³/h, air flow rate = 0 NL/min). These

data were fitted to the first and second order kinetic models, from the plot in Figure 6 (a), the reaction follows a first order reaction.

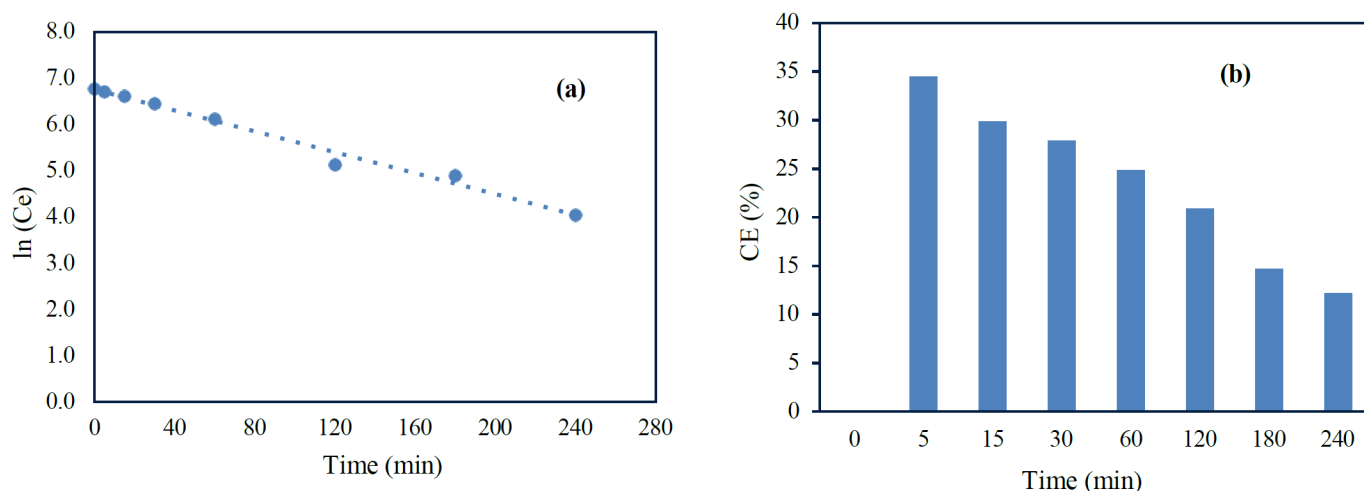


Figure 6—Kinetics (a) pseudo-first order model, (b) current efficiency (%)

The current efficiency with time is shown in Figure 6 (b), similar results to the PW were observed, the beginning of the reaction was determined to have the fastest COD decay kinetics. In contrast, it took 4 hours of treatment to reach the maximum removal with a 93% COD elimination rate as presented in Figure 6 (b). This observation can be explained by the mass transfer phenomenon. Initially, when the treatment starts, COD concentration near the electrode is high, which enhances the removal rate and leads to greater current efficiency. As the treatment progresses, the COD concentration gradually decreases, causing the mass transfer rate to decline, thus reducing the current efficiency.

Conclusion

This study assessed the treatment of synthetic PW and authentic SC wastewater streams using an advanced electrochemical oxidation method with a boron-doped diamond (BDD) electrode as the anode in a pilot-scale unit. Treatment of the synthetic PW demonstrated significant TOC reduction, with a maximum removal efficiency of 84% under optimized conditions (pH 12, current density of 200 mA/cm², and a circulation rate of 0.2 m³/hr). The influence of operating parameters such as current density, pH, and airflow on TOC removal was examined. Results indicated that higher pH values and increased current density substantially improved TOC removal efficiency, whereas airflow showed minimal impact. Similarly, the advanced oxidation process exhibited excellent performance in treating authentic SC, achieving a maximum COD removal of 91% in 3 hours under optimal conditions (pH 4, current density 150 mA/cm², and circulation rate of 0.2 m³/hr). These outcomes highlight the effectiveness and potential of advanced electrochemical treatment, particularly using the BDD anode, for managing the challenges associated with PW and SC wastewater. Further studies should focus on optimizing operational parameters to achieve even greater treatment efficiencies.

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